



RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Chapter S1B: Probability

SYLLABUS INCLUDES

- addition and multiplication of probabilities
- mutually exclusive events and independent events
- use of tables of outcomes, Venn diagrams, tree diagrams, and permutation and combination techniques to calculate probabilities
- calculation of conditional probabilities in simple cases
- use of:

$$P(A') = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

PRE-REQUISITES

- Basic Set Theory
- Permutations and Combinations

CONTENT

- 1 Random Experiments, Sample Space and Events**
 - 1.1 Definitions
 - 1.2 Set Operations and Properties involving Sets (Recap)
- 2 Probability**
 - 2.1 Theoretical Approach
 - 2.2 Properties involving Probabilities
 - 2.3 Probability Distribution
- 3 Mutually Exclusive Events**
- 4 Conditional Probability**
- 5 Independent Events**
 - 5.1 Definition
- 6 Probability Tree (Tree Diagram)**
- 7 Miscellaneous Examples**

Appendix A & B

INTRODUCTION

Perhaps it was man's thirst for gambling that led to the early development of probability theory. In an effort to increase their winnings, gamblers called upon great mathematicians such as Pascal, Leibniz, Fermat and James Bernoulli, to provide the optimum strategies for various games of chance. Gradually, these early developments branched out far beyond the games of chance to encompass many other fields associated with chance occurrences, such as politics, business, insurance, weather forecasting, and scientific research. For these predictions and generalizations to be fairly accurate, an understanding of basic probability theory is essential.

1 RANDOM EXPERIMENTS, SAMPLE SPACE AND EVENTS

1.1 Definitions

A **random experiment** is one whereby its outcome cannot be predicted. For example, throwing a die is a random experiment since the outcome of the throw cannot be predicted exactly.

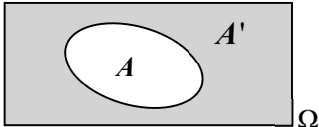
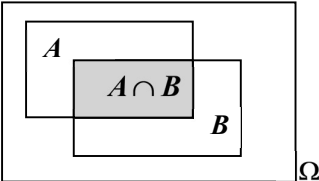
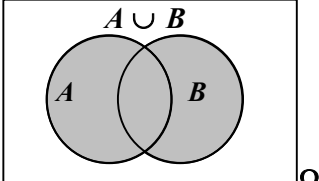
For each experiment, the set containing all possible outcomes is called the **sample space** Ω of the experiment. Each outcome of the experiment (i.e. each element in the sample space) is called a **sample point**. When we throw a die, the possible outcomes are the numbers 1, 2, 3, 4, 5 or 6. Hence the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$, and the six sample points are 1, 2, 3, 4, 5 and 6.

A subset of the sample space is called an **event**. For example, if we define A to be the event that an even number is obtained when we throw a die, then $A = \{2, 4, 6\}$. We say that A occurs when the outcome of the experiment is any element of the set A (i.e. any one of 2, 4 or 6 is obtained).

The event $\Omega = \{1, 2, 3, 4, 5, 6\}$ is certain to occur as the outcomes of the throw of a die must be one of the values 1, 2, 3, 4, 5 or 6. The empty set, denoted by $\{\}$ or $\emptyset$, is the event of **not** getting one of these values with the throw of a die and is called the impossible event.

1.2 Set Operations and Properties involving Sets (Recap)

Recall the following set operations:

Set Operation	Venn Diagram Representation
The complement of A , denoted by A' , is the event that <u>A does not occur</u> .	
The intersection of A and B , denoted by $A \cap B$, is the event that <u>A occurs and B occurs, i.e. both occur</u> . Note: A and B may occur simultaneously or in stages.	
The union of A and B , denoted by $A \cup B$, is the event that <u>A occurs or B occurs or both occur, i.e. at least one occurs</u> .	

For example, consider the experiment of throwing a die.

Let A be the event that an even number is obtained and B be the event that a prime number is obtained.

We have $\Omega = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 4, 6\}$ and $B = \{2, 3, 5\}$.

Then $A' = \{1, 3, 5\}$, $A \cap B = \{2\}$ and $A \cup B = \{2, 3, 4, 5, 6\}$.

Now, recall the following set properties:

If $n(A)$ and $n(B)$ denote the number of elements in the sets A and B respectively, where $A, B \subseteq \Omega$, then

- (i) $n(\emptyset) = 0$
- (ii) $n(\emptyset) \leq n(A) \leq n(\Omega)$
- (iii) If A is a subset of B , then $n(A) \leq n(B)$
- (iv) $n(A) + n(A') = n(\Omega)$
- (v) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- (vi) $n(A) = n(A \cap B) + n(A \cap B')$

2 PROBABILITY

Probability is a measure of the likelihood of an event occurring. An event that is more likely to occur has a higher probability than one that is less likely to occur. The probability of an event lies between 0 and 1 inclusively. An impossible event has a probability of zero and a sure event has a probability of 1. There are three approaches to assign a probability to a given event. These approaches are very different in principle and in practice.

2.1 Theoretical Approach (See Appendix A for 2 other approaches)

Consider the experiment of throwing a die. If the die is fair (i.e. unbiased), each of the six faces has an equal chance of turning up. That is to say that it is equally likely for the outcome to be any one of the six sample points 1, 2, 3, 4, 5 or 6.

We can therefore deduce that the probability of obtaining a '4' is 1 out of 6, i.e. $\frac{1}{6}$.

Similarly, the probability of obtaining an even number is 3 out of 6, i.e. $\frac{1}{2}$.

In doing so, we have assigned probability based on the theoretical assumptions about a fair die. When each of the sample point is equally likely to occur, we can assign probability in the following way:

If the sample space Ω consists of a finite number of equally likely outcomes, the probability of an event E is given by $P(E) = \frac{n(E)}{n(\Omega)}$, where $n(E)$ denotes the number of elements in the set E and $n(\Omega)$ denotes the number of elements in the set Ω .

i.e. Each card has an equal chance of being drawn.

Example 1

A card is drawn at random from an ordinary pack of 52 cards. Find the probability that the card is

- (i) a '7', (ii) neither an Ace nor a Heart.

Solution:

A pack of cards consists of 4 suits: spades, hearts, diamonds and clubs.
Each suit contains 13 cards:
2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A.

(i) $P(\text{drawing a '7'}) = \frac{4}{52} = \frac{1}{13}$.

(ii) $P(\text{neither an Ace nor a Heart}) = \frac{36}{52} = \frac{9}{13}$.

2.2 Properties involving Probabilities

Let Ω be the sample space of a random experiment and A and B be events where $A, B \subseteq \Omega$. We have the following properties (Compare with the set properties in Section 1.2):

- | | |
|-------|---|
| (i) | $P(\emptyset) = 0$ and $P(\Omega) = 1$ |
| (ii) | $0 \leq P(A) \leq 1$ |
| (iii) | If A is a subset of B , then $P(A) \leq P(B)$ |
| (iv) | $P(A) + P(A') = 1$ |
| (v) | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ |
| (vi) | $P(A) = P(A \cap B) + P(A \cap B')$ |
| (vii) | $P(A \cap B) = P(A) \times P(B A)$ (to be discussed later in Section 4) |

Example 2

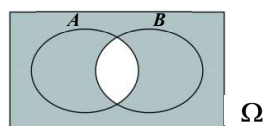
The events A and B are such that $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cap B) = 0.12$. Find the probability that

- | | |
|--|--|
| (i) at most 1 of the events A or B occurs, | (iii) either A or B occurs but not both, |
| (ii) at least 1 of the events A or B occurs, | (iv) neither A nor B occurs. |

Solution:

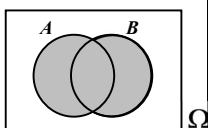
- (i) P (at most 1 of the events A or B occurs)

$$\begin{aligned}
 &= 1 - P(A \cap B) \\
 &= 1 - 0.12 \\
 &= 0.88.
 \end{aligned}$$



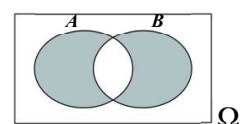
- (ii) P (at least 1 of the events A or B occurs)

$$\begin{aligned}
 &= P(A \cup B) \\
 &= P(A) + P(B) - P(A \cap B) \\
 &= 0.3 + 0.4 - 0.12 \\
 &= 0.58.
 \end{aligned}$$



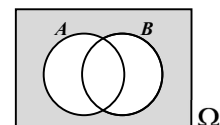
- (iii) P (either A or B occurs but not both)

$$\begin{aligned}
 &= P(A \cup B) - P(A \cap B) \\
 &= 0.58 - 0.12 \\
 &= 0.46.
 \end{aligned}$$



- (iv) P (neither A nor B occurs)

$$\begin{aligned}
 &= 1 - P(A \cup B) \\
 &= 1 - 0.58 \\
 &= 0.42.
 \end{aligned}$$



2.3 Probability Distribution

A **probability distribution** (usually in the form of a table) provides information on all possible outcomes of an experiment and the probability associated to each outcome.

Example 3

Two fair dice are thrown and T is the total score. Tabulate the probability distribution of T .

Solution:

T can take integer values 2 to 12.

$d_1 \backslash d_2$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Since each outcome is equally likely to occur, $P(T=2) = \frac{1}{36}$, $P(T=3) = \frac{2}{36} = \frac{1}{18}$ etc.

The probability distribution of T is as follows:

Total score	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$

Note: Total probability of all the possible outcomes is equal to 1.

Remark:

We will learn more about probability distribution table in Chapter S2A.

In reality, when considering probabilities involving experiments or random events, we may come across some information which may affect the values of these probabilities. For instance, the probability of a student not coming to school on a certain day is higher if he or she has not been well the day before. This concept is known as *conditional probability*, and there are many interesting related problems in real life, such as the *Monty Hall problem*.

4 CONDITIONAL PROBABILITY

Let A and B be events with $P(B) \neq 0$.

The probability of A occurring given that B has occurred, is given by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

This result may be written as
 $P(A \cap B) = P(B) \times P(A|B)$

To illustrate the concept of conditional probability, consider the following example:

Suppose two fair coins (labelled 1 and 2) are tossed and you are asked to place a bet on the outcome of the experiment. The sample space $\Omega = \{HH, HT, TH, TT\}$, and each of the four sample points has an equal chance of occurring, i.e. $\frac{1}{4}$.

If you bet that both the first and second tosses have the same outcome, i.e. “HH” or “TT”, your chance of winning the bet is 2 out of 4, i.e. $\frac{1}{2}$.

- (a) Now, if you are told that *there is at least one “H”*, what is the probability that you win the bet? Is it still $\frac{1}{2}$?

Reduced Sample Space

Since there is at least one “H”, the sample space is no longer $\{HH, HT, TH, TT\}$, but the **reduced sample space** $\{HH, HT, TH\}$, where each of the three sample points has an equal chance of occurring, i.e. $\frac{1}{3}$.

Let A be the event that both the first and second tosses have the same outcome.

Let B be the event that there is at least one “H”.

Then $A = \{HH, TT\}$, $B = \{HH, HT, TH\}$ and $A \cap B = \{HH\}$.

For you to win the bet, the only favourable outcome is “HH” since knowing that there is at least one “H” has ruled out “TT”.

The probability of winning the bet given that there is at least one “H” is 1 out of 3,

$$\text{i.e. } \frac{1}{3} = \frac{n(A \cap B)}{n(B)} = \frac{\frac{n(A \cap B)}{n(\Omega)}}{\frac{n(B)}{n(\Omega)}} = \frac{P(A \cap B)}{P(B)} = P(A|B)$$

which coincides with the formula of conditional probability.

- (b) What happens if you are told that the *first coin* shows “H” instead? Will your probability of winning the bet change again?

Let C be the event that the first coin shows “H”, i.e. $C = \{HH, HT\}$, which is another reduced sample space. Since there is one favourable outcome (i.e. “HH”) out of 2 possible outcomes, $P(A|C) = \frac{1}{2}$.

Example 5

A card is taken at random from an ordinary pack of 52 playing cards. A is the event that a Heart is taken, and B is the event that a King, Queen or Jack is taken. Find

- (i) $P(A|B)$, (ii) $P(B|A)$.

Solution:

We have $P(A) = \frac{13}{52}$, $P(B) = \frac{12}{52}$ and $P(A \cap B) = \frac{3}{52}$.

$$\begin{aligned} \text{(i)} \quad P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{3}{52} \div \frac{12}{52} = \frac{1}{4}. \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(B|A) &= \frac{P(B \cap A)}{P(A)} \\ &= \frac{3}{52} \div \frac{13}{52} = \frac{3}{13}. \end{aligned}$$

- (i) Alternatively, consider the reduced sample space:

Given that a King, Queen or Jack is taken, the reduced sample space consists of 12 cards. Out of these 12 cards, 3 of them are Hearts.

$$\text{Hence } P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{3}{12} = \frac{1}{4}.$$

- (ii) Alternatively, consider the reduced sample space:

Given that a Heart is taken, the reduced sample space consists of 13 cards. Of these 13 cards, 3 of them are King, Queen and Jack.

$$\text{Hence } P(B|A) = \frac{n(B \cap A)}{n(A)} = \frac{3}{13}.$$

Example 6

A fair die is thrown, and if the result is 2, 3, 4 or 5, that result is the score. If the result is 1 or 6, the die is thrown a second time and the sum of the two numbers resulting from both throws is the score. Events A and B are defined as follows:

A : the score is 5, 6, 7, 8 or 9,

B : the die is thrown two times.

Find (i) $P(A|B)$, (ii) $P(A)$, (iii) $P(B|A')$.

Solution:

Throw 1 Throw 2	1	2	3	4	5	6
1	2	2	3	4	5	7
2	3	2	3	4	5	8
3	4	2	3	4	5	9
4	5	2	3	4	5	10
5	6	2	3	4	5	11
6	7	2	3	4	5	12

(i) $P(A|B)$

$$= \frac{P(A \cap B)}{P(B)}$$

$$= \frac{P(\text{1st throw is 1 and 2nd throw is 4, 5 or 6}) + P(\text{1st throw is 6 and 2nd throw is 1, 2 or 3})}{P(\text{1st throw is 1 or 6})}$$

$$= \frac{\frac{3}{36} + \frac{3}{36}}{\frac{2}{6}} = \frac{\frac{1}{6}}{\frac{2}{6}} = \frac{1}{2}.$$

If the actions are mutually exclusive cases, we add.

If the actions are performed in succession, we multiply.

(ii) $P(A) = P(A \cap B) + P(A \cap B') = \frac{1}{6} + P(\text{1st throw is 5}) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$

(iii) $P(B|A') = \frac{P(B \cap A')}{P(A')} = \frac{P(B) - P(B \cap A)}{1 - P(A)} = \frac{\frac{2}{6} - \frac{1}{6}}{1 - \frac{1}{3}} = \frac{1}{4}.$

Note that answers for (ii) and (iii) can also be obtained using the table.

5 INDEPENDENT EVENTS

5.1 Definition

Two events A and B are **independent** if the probability of event A is unchanged by the occurrence of event B .

For example, if a coin is tossed twice, and A is the event that a “H” is shown in the first toss while B is the event that a “H” is shown in the second toss, it can be shown that A and B are independent.

From the definition, if A and B are 2 independent events, we have

$$\begin{aligned} P(A|B) &= P(A) \\ \Rightarrow \frac{P(A \cap B)}{P(B)} &= P(A) \\ \Rightarrow P(A \cap B) &= P(A) \times P(B) \end{aligned}$$

Note :

1. If events A and B with $P(A) > 0$, $P(B) > 0$ are independent events, then $P(A \cap B) = P(A)P(B) > 0$, therefore A and B *cannot be* mutually exclusive.
2. What can we say about the independence of A and B if A and B are mutually exclusive with $P(A) > 0$, $P(B) > 0$?
3. The independence of the two events cannot be 'seen' on the Venn diagram.

Independence plays an important role in probability and statistics. We may use any of the 3 equations below to check for independence:

$$\begin{aligned} &A \text{ and } B \text{ are independent events} \\ \Leftrightarrow &P(A|B) = P(A) \\ \Leftrightarrow &P(B|A) = P(B) \\ \Leftrightarrow &P(A \cap B) = P(A) \times P(B) \end{aligned}$$

Exercise: Show that if A and B are independent, the following events will also be independent.

- (a) A' and B (b) A and B' (c) A' and B'

We will show (a), ie A' and B are independent if A and B are independent.

Solution:

$$\begin{aligned} P(A' \cap B) &= P(B) - P(A \cap B) \\ &= P(B) - P(A)P(B) \text{ since } A \text{ and } B \text{ are independent} \\ &= P(B)[1 - P(A)] \\ &= P(B)P(A') \text{ (shown)} \end{aligned}$$

(b) and (c) can be proven in a similar way.

Example 7 (Same Context as Example 4)

A fair die is thrown three times. Events A , B and C are defined as follows:

- A - the total score is an odd number,
 B - a six appears at the first throw,
 C - the numbers that appear in the 3 throws are all even.

Determine if the following pairs of events are independent, stating your reasons clearly:

- (i) A and B , (ii) A and C , (iii) B and C

Solution:

$$P(A) = P(\text{all 3 odd}) + P(2 \text{ even and 1 odd}) = \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right) \times \frac{3!}{2!} = \frac{1}{2}$$

$$P(B) = \frac{1}{6}$$

$$P(C) = P(\text{all three throws are even}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$(i) \quad P(A \cap B) = P(1^{\text{st}} \text{ throw is 6 and total score odd}) = \frac{1}{6} \times \left(\frac{1}{2} \times \frac{1}{2} \times 2!\right) = \frac{1}{12}$$

$$P(A) \times P(B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

Since $P(A \cap B) = P(A) \times P(B)$, A and B are independent events.

OR: By considering the reduced sample space:

$$P(A | B) = P(2^{\text{nd}} \text{ and } 3^{\text{rd}} \text{ throws result in 1 odd and 1 even}) = \frac{1}{2} \times \frac{1}{2} \times 2! = \frac{1}{2}$$

Since $P(A | B) = P(A)$, A and B are independent events.

- (ii) From Example 4, A and C are mutually exclusive which implies $P(A \cap C) = 0$.

Since $P(A) > 0$, $P(C) > 0$, $P(A \cap C) \neq P(A) P(C)$

$\therefore A$ and C are not independent events.

$$(iii) \quad P(B \cap C) = P(1^{\text{st}} \text{ throw is 6 and all 3 throws are even}) = \frac{1}{6} \times \left(\frac{1}{2} \times \frac{1}{2}\right) = \frac{1}{24}$$

$$P(B) \times P(C) = \frac{1}{6} \times \frac{1}{8} = \frac{1}{48}$$

Since $P(B \cap C) \neq P(B) \times P(C)$, B and C are not independent events.

Example 8 [9740/2010/02/Q7]

For events A and B it is given that $P(A) = 0.7$, $P(B) = 0.6$ and $P(A|B') = 0.8$. Find

(i) $P(A \cap B')$, [2]

(ii) $P(A \cup B)$, [2]

(iii) $P(B'|A)$. [2]

For a third event C , it is given that $P(C) = 0.5$ and that A and C are independent.

(iv) Find $P(A' \cap C)$. [2]

(v) Hence state an inequality satisfied by $P(A' \cap B \cap C)$. [1]

Solution:

$$(i) \quad P(A|B') = \frac{P(A \cap B')}{P(B')}$$

$$\begin{aligned} P(A \cap B') &= P(A|B') \times P(B') \\ &= 0.8 \times (1 - 0.6) \\ &= 0.32 \end{aligned}$$

$$\begin{aligned} (ii) \quad P(A \cup B) &= P(A \cap B') + P(B) \\ &= 0.32 + 0.6 \\ &= 0.92 \end{aligned}$$

$$(iii) \quad P(B'|A) = \frac{P(B' \cap A)}{P(A)} = \frac{0.32}{0.7} = 0.457 \quad (3 \text{ s.f.})$$

(iv) Since A and C are independent, A' and C are also independent.

$$\begin{aligned} \therefore P(A' \cap C) &= P(A') \times P(C) \\ &= (1 - 0.7) \times 0.5 = 0.15 \end{aligned}$$

(v) Since $A' \cap B \cap C \subseteq A' \cap C$,

$$P(A' \cap B \cap C) \leq P(A' \cap C) = 0.15$$

$$\therefore P(A' \cap B \cap C) \leq 0.15$$

6 PROBABILITY TREE (TREE DIAGRAM)

A useful way of tackling many probability problems is to draw a tree diagram (or probability tree).

This method is illustrated in Examples 9 and 10.

Example 9

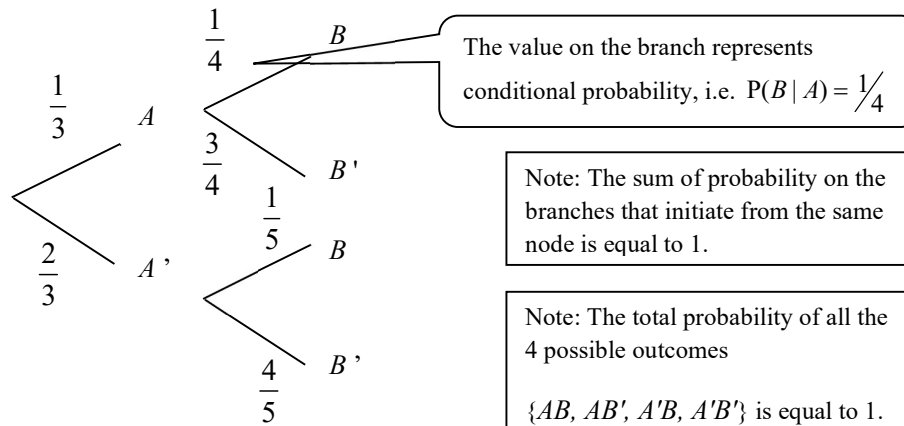
Two events A and B are such that $P(A) = \frac{1}{3}$, $P(B|A) = \frac{1}{4}$ and $P(B'|A') = \frac{4}{5}$.

By drawing a tree diagram, find

- (i) $P(B'|A)$, (ii) $P(A \cap B)$, (iii) $P(B)$, (iv) $P(A \cup B)$.

Solution:

Given $P(A) = \frac{1}{3}$, $P(B|A) = \frac{1}{4}$, $P(B'|A') = \frac{4}{5}$, construct the probability tree:



From the tree diagram,

(i) $P(B'|A) = \frac{3}{4}$

This follows from property (vii) in 2.2.

(ii) $P(A \cap B) = P(A) \times P(B|A) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$

(iii) $P(B) = P(A \cap B) + P(A' \cap B)$ Property (vi) in 2.2.

$$= \left(\frac{1}{3} \times \frac{1}{4} \right) + \left(\frac{2}{3} \times \frac{1}{5} \right) = \frac{13}{60}$$

(iv) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ Property (v) in 2.2.

$$= \frac{1}{3} + \frac{13}{60} - \frac{1}{12} = \frac{7}{15}$$

Example 10

The probability that a golfer hits the ball onto the green if it is windy as he strikes the ball is 0.4, and the probability that he hits the ball onto the green if it is not windy as he strikes the ball is 0.7. The probability that the wind will blow as he strikes the ball is 0.3.

Find the probability that

- (i) he hits the ball onto the green,
- (ii) it was not windy, given that he does not hit the ball onto the green.

Solution:

Let W denote the event that it is windy as he strikes the ball.

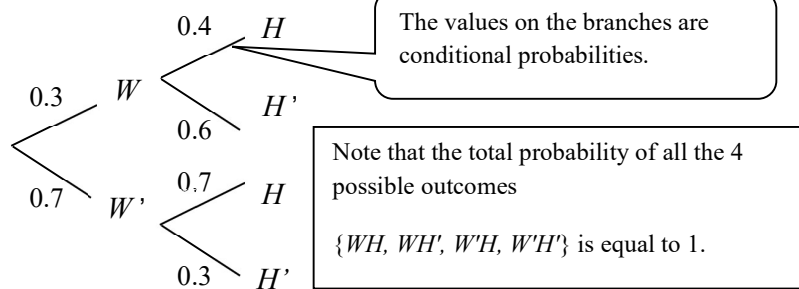
Let H denote the event that the golfer hits the ball onto the green.

Given

$$P(W) = 0.3$$

$$P(H|W) = 0.4$$

$$P(H|W') = 0.7$$



- (i) $P(\text{golfer hits the ball onto the green})$
 $= P(H)$
 $= P(W \cap H) + P(W' \cap H)$
 $= 0.3 \times 0.4 + 0.7 \times 0.7$
 $= 0.61$
- (ii) $P(\text{not windy} | \text{golfer does not hit the ball onto the green})$
 $= P(W' | H')$
 $= \frac{P(W' \cap H')}{P(H')}$
 $= \frac{0.7 \times 0.3}{1 - P(H)}$
 $= \frac{0.21}{0.39}$
 $= \frac{7}{13}$

Some results (See Appendix B for the proof)

- $P(A) \times P(B | A) = P(B) \times P(A | B) = P(A \cap B)$
- $P(B) = P(A \cap B) + P(A' \cap B) = P(A) \times P(B | A) + P(A') \times P(B | A')$
(Law of Total Probability)

7 MISCELLANEOUS EXAMPLES

Example 11

A child has a bag containing 12 sweets of which 3 are yellow, 5 are green and 4 are red. When the child wants to eat one of the sweets, a random selection is made from the bag and the chosen sweet is then eaten before the next random selection is made.

Find the probability that

- (i) the child selects a yellow and a green sweet in the first two selections,
- (ii) there is at least one yellow sweet in the first two selections,
- (iii) the fourth sweet selected is yellow, given that the first two sweets selected were red ones.

Solution:

- (i) P (the child selects a yellow and a green sweet in the first two selections)

$$= \left(\frac{3}{12} \times \frac{5}{11} \right) \times 2!$$

$$= \frac{5}{22}$$

- (ii) P (at least one yellow sweet in first 2 selections)

$$= 1 - P(\text{no yellow sweets in first 2 selections})$$

Complement Method

$$= 1 - \left(\frac{9}{12} \times \frac{8}{11} \right)$$

$$= 1 - \frac{6}{11}$$

$$= \frac{5}{11}$$

- (iii) Given that the first two sweets are red, the reduced sample space consists of 10 sweets, of which 3 are yellow, 5 are green and 2 are red.

$$P(4^{\text{th}} \text{ sweet yellow} \mid \text{first 2 are red})$$

$$= P(Y_3 Y_4 \mid 1^{\text{st}} 2 \text{ red}) + P(G_3 Y_4 \mid 1^{\text{st}} 2 \text{ red}) + P(R_3 Y_4 \mid 1^{\text{st}} 2 \text{ red})$$

$$= \left(\frac{3}{10} \times \frac{2}{9} \right) + \left(\frac{5}{10} \times \frac{3}{9} \right) + \left(\frac{2}{10} \times \frac{3}{9} \right) = \frac{3}{10}$$

If the actions are mutually exclusive cases, we add.

If the actions are performed in succession, we multiply.

Alternatively,

$$P(4^{\text{th}} \text{ sweet is yellow} \mid \text{first 2 are red})$$

$$= \frac{P(4^{\text{th}} \text{ sweet yellow and first 2 red})}{P(\text{first 2 red})}$$

$$= \frac{P(R_1 R_2 Y_3 Y_4) + P(R_1 R_2 G_3 Y_4) + P(R_1 R_2 R_3 Y_4)}{P(R_1 R_2)} = \dots$$

Example 12 [NJCPrelim9740/2012/02/Q8 (modified)]

In a particular school, all students are required to have a first Co-Curriculum Activity (CCA), while it is optional to have a second CCA. For their first CCAs, 45% are in Sports, 40% are in Music and Dance, and the remaining are in Clubs and Societies. $100p\%$ of students whose first CCA is Sports also have a second CCA. 75% of students whose first CCA is Music and Dance has no second CCA. Half of the students whose first CCA is Clubs and Societies have a second CCA.

- (i) Write down the probability that a randomly selected student has a second CCA, given that his first CCA is Music and Dance. [1]

The probability of a randomly selected student having a second CCA is $\frac{13}{40}$.

- (ii) Find the exact value of p . [2]

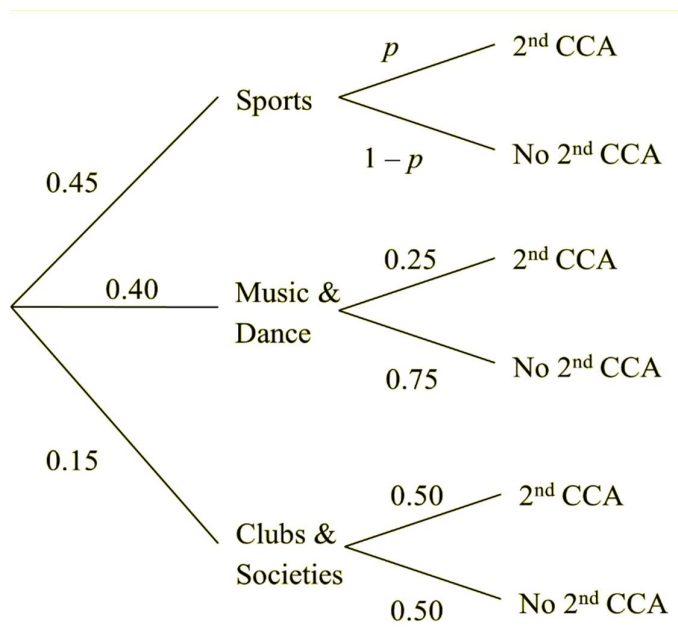
- (iii) Find the probability of a randomly selected student having Clubs and Societies as his first CCA, given that he does not have a second CCA. [3]

- (iv) Two students are selected at random. The events A and B are defined as follows.

A : one of the students has Music and Dance as first CCA and the other has Clubs and Societies as first CCA.

B : both students have a second CCA.

Find $P(A \cup B)$. [3]

Solution:

- (i) $P(\text{has second CCA} \mid \text{first CCA is Music and Dance}) = 0.25$

$$\begin{aligned}
 \text{(ii)} \quad P(\text{has second CCA}) &= 0.45p + 0.4(0.25) + 0.15(0.5) \\
 &= \frac{9p}{20} + \frac{7}{40}
 \end{aligned}$$

$$\frac{9p}{20} + \frac{7}{40} = \frac{13}{40}$$

$$p = \frac{1}{3}$$

$$\begin{aligned}
 \text{(iii)} \quad &P(\text{first CCA Clubs and Societies} \mid \text{no second CCA}) \\
 &= \frac{P(\text{first CCA Clubs and Societies and no second CCA})}{P(\text{no second CCA})}
 \end{aligned}$$

$$= \frac{0.15(0.5)}{1 - \frac{13}{40}}$$

$$= \frac{1}{9}$$

$$\begin{aligned}
 \text{(iv)} \quad P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
 &= (0.4)(0.15)2! + \left(\frac{13}{40}\right)^2 - [(0.4)(0.25)(0.15)(0.5)]2! \\
 &= \frac{337}{1600}
 \end{aligned}$$

Note :

Since 45% of the students are in Sports, 40% of them are in Music and Dance and 15% of them are in Clubs and Societies, the probability of selecting a student who is in Music and Dance is 0.4. However, after this student is selected, the probability of selecting the next student who is in Clubs and Societies may not be 0.15. Thus, for the computation in **(iv)** to hold, we assume that the number of students in the school is large enough so that the probabilities of selecting the next student in Sports, Music and Dance and Clubs and Societies remain approximately at 0.45, 0.4 and 0.15 respectively.

Example 13 [HCI JC2 CT2 9758/2017/P2/Q5] (modified)

A sports delegation comprising 5 male and 5 female athletes is shown in the table below.

	basketball	tennis	badminton	netball
Male	3	2	0	0
Female	1	1	2	1
Total	4	3	2	1

A committee of 4 athletes is to be formed from these 10 athletes. Find the probability that

- (i) the committee comprises 2 male and 2 female athletes such that both male athletes play the same sport while none of the female athletes plays the same sport as the male athletes,
- (ii) exactly 3 female athletes are chosen given that exactly 1 badminton player is chosen,
- (iii) the committee consists of exactly 1 badminton player or exactly 1 tennis player (or both).

Solution:

- (i) Required probability = P (2 male basketball) + P (2 male tennis)

$$= \left[\left(\frac{3}{10} \times \frac{2}{9} \times \frac{4}{8} \times \frac{3}{7} \right) + \left(\frac{2}{10} \times \frac{1}{9} \times \frac{4}{8} \times \frac{3}{7} \right) \right] \times \frac{4!}{2!2!} = \frac{4}{35}$$

OR: Required probability = $\frac{n(2 \text{ male basketball}) + n(2 \text{ male tennis})}{n(\text{committees without restriction})}$

$$= \frac{{}^3C_2 {}^4C_2 + {}^2C_2 {}^4C_2}{{}^{10}C_4} = \frac{4}{35}$$

- (ii) $\frac{P(\text{exactly 3 female} \mid \text{exactly 1 badminton})}{P(\text{exactly 1 badminton})}$

$$= \frac{P(1 \text{ female badminton, 2 other female, 1 male})}{P(\text{exactly 1 badminton})} = \frac{\frac{2}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{5}{7} \times \frac{4!}{2!}}{\frac{2}{10} \times \frac{8}{9} \times \frac{7}{8} \times \frac{6}{7} \times \frac{4!}{3!}} = \frac{15}{56}$$

OR: $\frac{P(\text{exactly 3 female} \mid \text{exactly 1 badminton})}{P(\text{exactly 1 badminton})}$

$$= \frac{n(1 \text{ female badminton, 2 other female, 1 male})}{n(\text{exactly 1 badminton})} = \frac{{}^2C_1 {}^3C_2 {}^5C_1}{{}^2C_1 {}^8C_3} = \frac{15}{56}$$

- (iii) Let A be the event that the committee contains exactly 1 badminton player, and B be the event that the committee contains exactly 1 tennis player.

$$P(A) = \left(\frac{2}{10} \times \frac{8}{9} \times \frac{7}{8} \times \frac{6}{7} \right) \times \frac{4!}{3!} = \frac{8}{15} \quad \text{OR} \quad P(A) = \frac{n(A)}{n(\Omega)} = \frac{{}^2C_1 {}^8C_3}{{}^{10}C_4}$$

$$P(B) = \left(\frac{3}{10} \times \frac{7}{9} \times \frac{6}{8} \times \frac{5}{7} \right) \times \frac{4!}{3!} = \frac{1}{2} \quad \text{OR} \quad P(B) = \frac{n(B)}{n(\Omega)} = \frac{{}^3C_1 {}^7C_3}{{}^{10}C_4}$$

$$P(A \cap B) = P(\text{exactly 1 badminton and exactly 1 tennis})$$

$$= \left(\frac{2}{10} \times \frac{3}{9} \times \frac{5}{8} \times \frac{4}{7} \right) \times \frac{4!}{2!} = \frac{2}{7} \quad \text{OR} \quad P(A \cap B) = \frac{n(A \cap B)}{n(\Omega)} = \frac{{}^2C_1 {}^3C_1 {}^5C_2}{{}^{10}C_4}$$

$$\therefore P(\text{exactly 1 badminton or exactly 1 tennis (or both)})$$

$$= P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B)$$

$$= \frac{8}{15} + \frac{1}{2} - \frac{2}{7}$$

$$= \frac{157}{210}$$

CONCLUSION

Probability is a measure of the likelihood that something will or will not occur. Our environment evokes the probability situation every day. As you constantly investigate the notion of fairness and chance, and understand all the possible outcomes of a particular event through the exposure to probability, you will gradually build the capacity to make informed decisions and opinions in your everyday life.

APPENDIX A

Other approaches to assign probability to a given event:

- **Empirical Approach**

Consider the experiment of throwing a die.

If the die is biased, can we determine the probability of obtaining a “4”?

Well, we can throw the die repeatedly, and keep track of the number of throws (n) and the number of times “4” appears (f). Then calculate the ratio $\frac{f}{n}$. As n increases, we can expect this ratio to stabilise.

The probability is the limiting value of this ratio as n tends to infinity. In this case, the probability is derived from empirical evidence.

In the long run, the ratio $\frac{f}{n}$ tends to the actual probability of the event.

This ratio is sometimes referred to as the **relative frequency** or **proportion**.

- **Subjective Approach**

Not everything in life can be repeated in practice.

For example, if you want to evaluate your chances of scoring an “A” in your first sitting of the GCE A-Level H2 Mathematics Examination, you have only one attempt. You can however make an educated guess about the probability based on your understanding of the concepts, the quality of your written work, the results of your common tests/exams, etc. Such a subjective approach to probability is just as important as the other 2 more systematic and logical approaches.

APPENDIX B

Prove the following results:

- (a) $P(A) \times P(B | A) = P(B) \times P(A | B) = P(A \cap B)$
 (b) $P(B) = P(A \cap B) + P(A' \cap B) = P(A) \times P(B | A) + P(A') \times P(B | A')$
(Law of Total Probability)

Proof:

- (a) Since $P(A | B) = \frac{P(A \cap B)}{P(B)}$ and $P(B | A) = \frac{P(B \cap A)}{P(A)}$,
 $P(A \cap B) = P(B) \times P(A | B)$ and $P(B \cap A) = P(A) \times P(B | A)$
 $\therefore P(A \cap B) = P(A)P(B | A) = P(B)P(A | B)$ (Proved)
- (b) We have $P(B) = P(B \cap A) + P(B \cap A') \dots\dots(I)$
 Since $P(B \cap A) = P(A) \times P(B | A) \dots\dots(II)$
 and $P(B \cap A') = P(A') \times P(B | A') \dots\dots(III)$,
 Substitute (II) and (III) into (I), the result is obtained.

SUMMARY